

SOFTWARE REQUIREMENTS SPECIFICATION

Design and Development of a Web-Based Tourism Platform for Exploring Morocco

— *Morocco Explore* —

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Project Type: Personal Project

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of **Morocco Explore**, a web-based tourism information platform dedicated to helping users discover tourist destinations in Morocco.

This document describes the objectives, scope, features, users, constraints, technical requirements, and expected behavior of the system. It will serve as a reference throughout the different stages of the project, including analysis, design, development, testing, and deployment.

1.2 Project Overview

Morocco Explore is a web platform designed to provide users with information about tourist destinations in Morocco.

The platform allows visitors to explore different destinations, discover tourist attractions, view images and descriptions, search for specific places, and filter destinations according to different criteria.

Registered users can additionally save their favorite places, rate destinations, and share reviews.

The platform also includes an administration interface that allows the administrator to manage destinations, categories, users, and reviews.

The main objective is to provide a simple, attractive, and user-friendly platform that

helps people discover Morocco and plan their visits.

1.3 Scope of the Document

This SRS defines the requirements for the first version of Morocco Explore.

It covers:

- The different types of users.
- Destination management.
- Category management.
- Search and filtering.
- Destination information.
- Favorites.
- Ratings and reviews.
- User account management.
- Administration.
- Security and performance requirements.

The document will serve as a foundation for the system design and implementation phases.

1.4 Intended Audience

This document is intended for:

- Software developers.
- System designers.
- Software testers.
- Project contributors.
- Future maintainers.

- Anyone interested in understanding the system requirements.

1.5 Definitions and Acronyms

Term	Definition
Morocco Explore	The tourism web platform developed in this project.
Visitor	A person who accesses the platform without creating an account.
User	A registered person who can use additional platform features.
Administrator	The person responsible for managing the platform and its content.
Destination	A tourist place presented on the platform.
Category	A classification used to organize tourist destinations.
Review	A text comment submitted by a registered user about a destination.
Rating	A numerical evaluation given by a user to a destination.
Favorite	A destination saved by a user for later access.
SRS	Software Requirements Specification.
UI	User Interface.
DBMS	Database Management System.

2. Overall Description

2.1 Product Perspective

Morocco Explore is a web-based application that provides tourism-related information through a modern and responsive interface.

The system follows a client-server architecture. Users interact with the application through a web browser, while the server manages the application logic and communicates with the database.

The database stores information about destinations, categories, users, favorites, ratings, and reviews.

The platform is designed to be accessible from different devices, including desktop computers, tablets, and smartphones.

2.2 Product Objectives

The main objectives of Morocco Explore are:

- To provide an online platform for discovering tourist destinations in Morocco.
- To organize tourist destinations into categories.
- To provide detailed information about each destination.
- To facilitate destination discovery through search and filtering.
- To allow users to save their favorite destinations.
- To allow users to rate and review destinations.
- To provide administrators with tools for managing the platform.
- To offer a simple and attractive user experience.

2.3 Target Users

The platform targets:

- Tourists visiting Morocco.
- People planning a trip to Morocco.
- Moroccan residents looking for places to visit.
- Users interested in Moroccan culture, nature, history, and tourism.
- Anyone who wants to discover tourist destinations in Morocco

2.4 User Characteristics

Visitor

A visitor does not need an account to explore the platform.

The visitor can:

- Access the homepage.
- Browse destinations.
- Browse categories.
- Search for destinations.
- Filter destinations.
- View destination details.

Registered User

A registered user has access to additional features.

The user can:

- Manage their profile.
- Save destinations to favorites.

- Rate destinations.
- Write reviews.
- View their saved destinations.

Administrator

The administrator manages the platform.

The administrator can:

- Manage destinations.
- Manage categories.
- Manage users.
- Manage reviews.
- Manage ratings.
- View platform statistics.

2.5 Product Functions

The main functions of the platform are:

- User registration and authentication.
- Destination browsing.
- Destination search.
- Destination filtering.
- Destination details.
- Category browsing.
- Favorites management.
- Ratings and reviews.
- User profile management.
- Destination management.
- Category management.

- User management.
- Review moderation.

2.6 Operating Environment

Morocco Explore will operate as a web application.

Users will access the platform through modern web browsers such as:

- Google Chrome.
- Mozilla Firefox.
- Microsoft Edge.
- Safari.

The application will support:

- Desktop computers.
- Laptops.
- Tablets.
- Smartphones.

An Internet connection is required to access the online platform

2.7 Assumptions and Dependencies

The project assumes that:

- Users have access to the Internet.
- Users have access to a modern web browser.
- The administrator provides accurate destination information.
- Destination images and descriptions are available.
- The hosting server and database are operational.
- External map services, if used, remain available.

2.8 Constraints

The project has the following constraints:

- The application will be web-based.
- The platform must be responsive.
- User passwords must be securely stored.
- Only authorized administrators can manage platform content.
- Users must be authenticated to submit reviews and manage favorites.
- The system must protect user data from unauthorized access.

3. Functional Requirements

3.1 Visitor Access

Description

The system shall allow visitors to access the main tourism content without creating an account.

Requirements

The visitor shall be able to:

- Access the homepage.
- Browse destinations.
- Browse categories.
- Search for destinations.
- Filter destinations.
- View destination details.
- View destination images.
- View destination location information.

Authentication shall not be required for basic destination browsing.

3.2 User Registration and Authentication

Description

The system shall provide a secure authentication system for users.

Registration

The user shall be able to create an account using:

- Full name.
- Email address.
- Password.
- Password confirmation.

The system shall verify that the email address is unique.

Login

The user shall be able to log in using:

- Email address.
- Password.

The system shall verify the provided credentials.

Logout

The authenticated user shall be able to log out securely.

3.3 Destination Browsing

The system shall display a collection of tourist destinations.

Each destination may contain:

- Name.
- Description.
- Main image.
- Additional images.
- City or region.

- Category.
- Location.
- Opening information, if applicable.
- Additional useful information.

Users shall be able to browse the available destinations.

3.4 Destination Details

When a user selects a destination, the system shall display a detailed page.

The page shall include:

- Destination name.
- Description.
- Images.
- Location.
- Category.
- Rating.
- User reviews.
- Additional information.
- Map location, when available.

Example:

Ouzoud Waterfalls

Located in the Béni Mellal-Khénifra region, the Ouzoud Waterfalls are one of Morocco's most popular natural attractions.

3.5 Search Functionality

The system shall provide a search function allowing users to find destinations.

Users shall be able to search by:

- Destination name.
- City.
- Region.
- Category.

The system shall display destinations matching the search query.

3.6 Filtering and Sorting

The platform shall allow users to filter destinations.

Possible filters include:

- City.
- Region.
- Category.
- Rating.

Users may also sort results according to available criteria, such as:

- Alphabetical order.
- Highest rating.
- Most popular.

3.7 Favorites

Registered users shall be able to save destinations to a personal favorites list.

The user shall be able to:

- Add a destination to favorites.
- Remove a destination from favorites.

- View all saved destinations.

Visitors shall not have access to persistent favorites.

3.8 Reviews and Ratings

Registered users shall be able to evaluate destinations.

A user can:

- Give a rating.
- Write a review.
- Edit their review.
- Delete their review.

The system shall display approved reviews on the corresponding destination page.

The administrator shall have the ability to moderate reviews.

3.9 User Profile Management

Registered users shall have access to a personal profile.

The user shall be able to:

- View profile information.
- Edit personal information.
- Change their password.
- View favorite destinations.
- View their submitted reviews.

3.10 Administration

The platform shall provide a secure administration interface.

The administrator shall be able to manage:

- Destinations.
- Categories.
- Users.
- Reviews.

The administration interface shall only be accessible to authorized administrators.

3.11 Content Management

The administrator shall be able to create and manage tourist destinations.

For each destination, the administrator shall be able to:

- Add a destination.
- Edit destination information.
- Delete a destination.
- Add images.
- Assign a category.
- Specify the location.
- Publish or unpublish the destination.

The administrator shall also be able to manage categories.

3.12 User Management

The administrator shall be able to:

- View registered users.
- Search for users.
- View user information.
- Activate or deactivate user accounts.

- Delete user accounts when necessary.

3.13 Review Management

The administrator shall be able to:

- View submitted reviews.
- Approve reviews.
- Hide inappropriate reviews.
- Delete reviews when necessary.

This functionality helps maintain the quality and reliability of the information presented on the platform.

4. Non-Functional Requirements

4.1 Usability

The platform shall provide:

- A simple and intuitive interface.
- Clear navigation.
- Readable content.
- Consistent design.
- Easy access to destination information.

A new user should be able to browse the platform without requiring special instructions.

4.2 Performance

The system should provide fast response times for common operations such as:

- Loading destinations.
- Searching.
- Filtering.
- Viewing destination details.

Images should be optimized to reduce loading times.

4.3 Security

The system shall:

- Secure user authentication.
- Encrypt or hash user passwords using appropriate security mechanisms.

- Prevent unauthorized access to administrative features.
- Validate user input.
- Protect against common web vulnerabilities.
- Protect user data.

4.4 Reliability

The system should operate consistently and handle common user operations without unexpected failures.

Errors should be handled appropriately and informative error messages should be provided when necessary.

4.5 Availability

The platform should be available whenever the hosting infrastructure is operational.

The system should minimize downtime during normal operation.

4.6 Maintainability

The application should be developed using a clear and organized structure.

The code should be:

- Modular.
- Readable.
- Documented where necessary.
- Easy to modify.
- Easy to extend.

4.7 Scalability

The architecture should allow the platform to grow in the future.

It should be possible to add:

- More destinations.
- More categories.
- More users.
- More reviews.
- Additional tourism-related features.

without requiring a complete redesign of the system.

4.8 Responsiveness

The interface shall adapt to different screen sizes.

The platform should provide a good user experience on:

- Smartphones.
- Tablets.
- Laptops.
- Desktop computers.

4.9 Compatibility

The platform should be compatible with modern versions of major web browsers.

5. External Interface

Requirements

5.1 User Interface

The platform shall provide a modern and intuitive user interface.

The main pages may include:

Public pages

- Home
- Destinations
- Categories
- Destination Details
- Search Results
- About
- Contact
- Login
- Registration

User pages

- User Dashboard
- Profile
- Favorites
- My Reviews

Administration pages

- Admin Dashboard
- Destination Management
- Category Management
- User Management
- Review Management

5.2 Map Interface

The platform may integrate an online map service to display the geographic location of destinations.

The map should allow users to easily identify where a destination is located.

5.3 Database Interface

The application shall communicate with a relational database to store and retrieve:

- User data.
- Destination data.
- Category data.
- Review data.
- Rating data.
- Favorite data.

5.4 Communication Interface

The application will communicate between the client and server using standard web protocols such as HTTP/HTTPS.

HTTPS should be used in the production environment to secure communication.

6. Data Requirements

6.1 Main Entities

The first version of the platform is expected to contain the following main entities:

User

Stores registered user information.

Category

Represents a group of tourist destinations.

Examples:

- Historical Sites
- Nature
- Beaches
- Museums
- Cultural Places

Destination

Represents a tourist place.

Image

Stores images associated with destinations.

Favorite

Represents a user's saved destination.

Review

Stores reviews submitted by users.

Rating

Represents a user's rating of a destination.

6.2 Data Storage

The system shall use a relational database.

The database shall maintain relationships between:

- Users and favorites.
- Users and reviews.
- Users and ratings.
- Categories and destinations.
- Destinations and images.

The detailed database structure will be defined during the System Design phase.

6.3 Data Validation

The system shall validate data before storing it.

For example:

- Required fields must not be empty.
- Email addresses must have a valid format.
- Duplicate emails shall not be accepted.
- Ratings must respect the defined rating range.
- Uploaded images must respect allowed formats and size limits.

7. System Rules and Business Rules

The platform shall follow the following rules:

Rule 1 — Public Access

Visitors can browse tourist destinations without creating an account.

Rule 2 — Authentication

An account is required to use personalized features such as favorites and reviews.

Rule 3 — Reviews

Only authenticated users can submit reviews.

Rule 4 — Ratings

A user can rate a destination according to the defined rating system.

Rule 5 — Favorites

A user can save a destination only once in their favorites.

Rule 6 — Administration

Only the administrator can create, modify, or delete platform content.

Rule 7 — Review Moderation

The administrator can hide or delete inappropriate reviews.

Rule 8 — Destination Publication

Only published destinations are visible to normal visitors.

8. Technologies and Technical Requirements

The exact technologies can be finalized before the development phase.

A recommended technology stack for this project is:

Front-end

- HTML5
- CSS3
- JavaScript
- Bootstrap or Tailwind CSS

Back-end

Laravel / PHP

Database

MySQL

Development Tools

- Visual Studio Code
- Git
- GitHub
- MySQL / phpMyAdmin

Map Integration

A map service such as OpenStreetMap can be considered for displaying destination locations.

9. Project Scope

9.1 In Scope

The first version of Morocco Explore includes:

- Public access to destinations.
- Destination catalog.
- Destination categories.
- Destination details.
- Search.
- Filtering.
- User registration.
- User authentication.
- User profiles.
- Favorites.
- Ratings.
- Reviews.
- Administration dashboard.
- Destination management.
- Category management.
- User management.
- Review moderation.
- Responsive design.
- Map integration.

9.2 Out of Scope

The following features are not included in the first version:

- Hotel booking.
- Flight booking.
- Restaurant reservations.
- Online payment.
- Tour guide booking.
- Real-time chat.
- Live tourist assistance.
- Complete trip planning.
- AI-based travel recommendations.

These features may be considered as future enhancements.

10. Project Timeline

The project can be divided into several phases.

Phase	Main Activities
Phase 1	Requirements analysis and SRS
Phase 2	System analysis and UML
Phase 3	Database design
Phase 4	UI/UX design
Phase 5	Backend development
Phase 6	Frontend development
Phase 7	Integration
Phase 8	Testing and bug fixing
Phase 9	Deployment
Phase 10	Documentation and final improvements

11. Future Enhancements

Future versions of Morocco Explore may include:

- Personalized travel recommendations.
- Complete trip planning.
- Interactive itineraries.
- Weather information.
- Hotel information.
- Restaurant recommendations.
- Tourist activities.
- Multilingual support.
- Mobile application.
- Social sharing.
- Interactive maps with routes.
- AI-powered recommendations.
- User travel diaries.

12. Conclusion

Morocco Explore is a web-based tourism platform designed to facilitate the discovery of tourist destinations in Morocco.

The platform provides visitors with accessible information about destinations and allows registered users to personalize their experience through favorites, ratings, and reviews. An administration interface ensures that the platform's content can be efficiently managed.

This Software Requirements Specification establishes the foundation for the development of the platform. The next stage will focus on analyzing these requirements and designing the system architecture, database, and UML diagrams.